



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

18-05-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PART-1_PHYSICS

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	C	D	B	C	C	B
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	B,D	A,C	A,B,C,D	B,D	A,C	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	8.00	2.00	5.00	8.00	2.00	17.00

PART-2_CHEMISTRY

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	C	C	A	B	C	A
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	B,D	B,C,D	A,B,C	A,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	3.00	4.50	3.00	6.00	556.10 to 557.80	8.90 to 9.10

PART-3_MATHEMATICS

SECTION-I(i)	Q.	1	2	3	4	5	6
	A.	D	C	D	A	B	B
SECTION-I(ii)	Q.	7	8	9	10	11	12
	A.	B,C,D	A,C,D	A,B,D	C,D	A,B,C	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	9.50	119.00	0.50	1206.00	19.20	5.00

HINT – SHEET

PART-1_PHYSICS

SECTION-I(i)

1. Ans (C)

$$A_r = \frac{\sqrt{\mu_1} - \sqrt{\mu_2}}{\sqrt{\mu_1} + \sqrt{\mu_2}} A_i$$

$$A_t = \frac{2\sqrt{\mu_1}}{\sqrt{\mu_1} + \sqrt{\mu_2}} A_i$$

$$\Rightarrow \frac{A_r}{A_t} = -\frac{1}{2} \text{ and } P = \frac{1}{2} \mu v \omega^2 A^2$$

$$= \frac{1}{2} \mu \sqrt{\frac{T}{\mu}} v \omega^2 A^2 \quad \left(\because v = \sqrt{\frac{T}{\mu}} \right)$$

$$P \propto \sqrt{\mu} A^2$$

$$\left[\frac{P_t}{P_r} = \sqrt{\frac{\mu_t}{\mu_r}} \left(\frac{A_t}{A_r} \right)^2 \right]$$

$$= \sqrt{\frac{4\mu}{\mu}} (2)^2 = 8$$

2. Ans (D)

$$2R = \frac{3R}{2} + \frac{PdV}{dT}$$

$$\frac{PdV}{dT} = \frac{R}{2} \Rightarrow \frac{RT}{V} \frac{dV}{dT} = \frac{R}{2}$$

$$\Rightarrow \int \frac{dV}{V} = \frac{1}{2} \int \frac{dT}{T}$$

$$\ln V + \ln C = \frac{1}{2} \ln T \text{ (where C is constant)}$$

$$\ln CV = \frac{1}{2} \ln T \Rightarrow T = C^2 V^2$$

$$\text{Hence } T \propto V^2$$

$\therefore$ Temperature becomes four times.

3. Ans (B)

$$I_0 = \frac{1}{4} \times (4M) \frac{a^2}{6}$$

$$= \frac{Ma^2}{6} = \frac{7 \times 144}{6} = 168 \text{ kg cm}^2$$

4. **Ans (C)**

Lens formula gives

$$\frac{1}{v} - \frac{1}{(-100)} = \frac{1}{50} \Rightarrow v = 100 \text{ cm}$$

The mirror forms an image 50 cm behind itself and the image is moving to right with a velocity of 20 m/s. The image formed by the mirror will act as moving object for the lens.

Again applying the lens formula gives

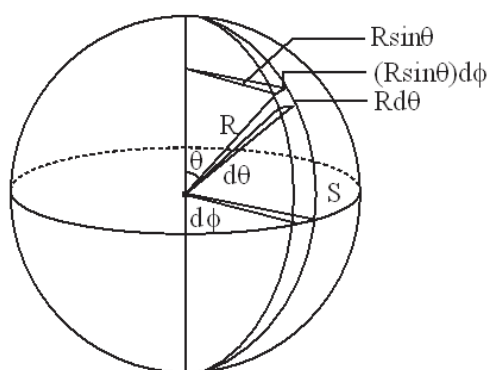
$$\frac{1}{v} - \frac{1}{(-200)} = \frac{1}{50} \Rightarrow v = \frac{200}{3} \text{ cm}$$

For lens

$$\frac{dv}{dt} = \left(\frac{v}{u}\right)^2 \frac{du}{dt} = \left(\frac{200/3}{200}\right)^2 \cdot 20 = \frac{20}{9} \text{ cm/s}$$

5. **Ans (C)**

The given charge distribution is equivalent to charge on a strip S on a uniformly charged spherical shell of radius R.



Since electric field in the sphere is zero, therefore net electric field due to any two diametrically opposite such strips on all points of the common diameter must be zero. It suggests that electric field of a single strip and hence electric field of the given semicircular arc anywhere on their diameter cannot have a component parallel to the diameter.

6. **Ans (B)**

$$\frac{dN}{dt} = t^2 - \lambda N$$

$$\text{For } \frac{dN}{dt} \text{ to be minimum } \frac{d^2N}{dt^2} = 0$$

$$\Rightarrow \frac{d^2N}{dt^2} = 2t - \lambda(t^2 - \lambda N) = 0$$

$$\text{Or } N = \frac{2t_0 - \lambda t_0^2}{\lambda^2}$$

PART-1_PHYSICS

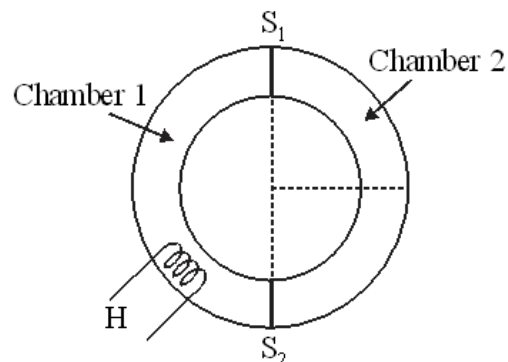
SECTION-I(ii)

7. **Ans (B,D)**

The process in chamber 2 is adiabatic.

Let final pressure in chamber 2 be P_2 .

$$P_2 \left(\frac{V_0}{2}\right)^\gamma = P_0 V_0^\gamma$$



$$\Rightarrow P_2 = (2)^{3/2} P_0 \left[\because \gamma = \frac{3}{2} \right]$$

$$= 2\sqrt{2} P_0$$

In equilibrium, pressure in both chambers must be same, hence final pressure in chamber 1 is

$$P_1 = P_2 = 2\sqrt{2} P_0$$

Work done on the gas in chamber 2 by gas in chamber 1 is

$$W = \frac{P_2 V_2 - P_0 V_0}{\gamma - 1} = \frac{2\sqrt{2} \cdot P_0 \frac{V_0}{2} - P_0 V_0}{\frac{3}{2} - 1}$$

$$W = 2(\sqrt{2} - 1) P_0 V_0$$

Change in internal energy of the gas in chamber 1

$$\Delta U = n C_v \Delta T = \frac{n R \Delta T}{\gamma - 1} = 2n R \Delta T$$

$$= 2[P_1 V_1 - P_0 V_0] = 2 \left[2\sqrt{2} P_0 \cdot \frac{3}{2} V_0 - P_0 V_0 \right]$$

$$= 2(3\sqrt{2} - 1) P_0 V_0$$

$$\therefore \Delta Q = \Delta U + W = 2(3\sqrt{2} - 1) P_0 V_0 + 2(\sqrt{2} - 1) P_0 V_0$$

$$= 4P_0 V_0 [2\sqrt{2} - 1]$$

8. Ans (A,C)

Loss in electrostatic potential energy

$$U_i - U_f = \lambda E_0 \frac{L^2}{2}$$

This is equal to gain in gravitational potential energy

$$\lambda E_0 \frac{L^2}{2} = Mg \frac{L}{2} \Rightarrow E_0 = \frac{Mg}{\lambda L}$$

The electric force on the rod is $F_e = \lambda L E_0$.

Force can be considered at centre for writing the torque.

In equilibrium,

$$\tau_{Fe} = \tau_{mg} \text{ [torque about hinge]}$$

$$\lambda L E_0 \frac{L}{2} \cos \theta = Mg \frac{L}{2} \sin \theta$$

$$\text{Putting } E_0 = \frac{Mg}{\lambda L} \text{ we get } \theta = 45^\circ$$

The rod has maximum KE at this position.

9. Ans (A,B,C,D)

$$mg \left(h + \frac{H}{2} \right) = [\nu \rho g] \frac{H}{2}$$

$$h = H$$

$$mg \left(h + \frac{H}{2} + \frac{H}{2} \right) = [\nu \rho g] \frac{H}{2} + [\nu 2 \rho g] \frac{H}{2}$$

$$h = \frac{7H}{2}$$

$$mg \left(h + \frac{H}{2} \right) - [\nu \rho g] \frac{H}{2} = K.E.$$

$$mg \left(\frac{5H}{2} \right) - \frac{3}{2} mgH = K.E.$$

$$K.E. = mgH.$$

10. Ans (B,D)

$$W_g + W_{fr} = \Delta K$$

W_g and ΔK same for all cases so W_{fr} should also be same

$$N_1 < N_2 \text{ and } \ell_1 < \ell_2 \Rightarrow \mu_1 > \mu_2$$

[N = Normal forces, ℓ = length of the path]

$$N_2 > N_3 \text{ and } \ell_2 = \ell_3 \Rightarrow \mu_2 < \mu_3$$

11. Ans (A,C)

For $r = 0$

$$R_{eq} = 1.5 R$$

for $r = \infty$

$$R_{eq} = 2R.$$

12. Ans (A,C,D)

Consider the X and Y coordinate axes along AB and BC respectively.

Time taken to collide BC is given by

$$0 = \frac{ev\sqrt{3}}{2} - \frac{g\sqrt{3}}{2} t$$

$$\Rightarrow t = \frac{ev}{g}$$

In x-direction

$$2h = \left(\frac{v}{2} \right) t + \frac{1}{2} \left(\frac{g}{2} \right) t^2$$

$$\text{or } e^2 + 2e - \frac{8gh}{v^2} = 0$$

$$\therefore e = \frac{1}{\sqrt{3}}$$

$$(B) t = \frac{eV}{g} = \sqrt{\frac{8}{g}} s$$

$$(C) v_x = \frac{v}{2} + \frac{g}{2} t = \frac{v}{2} (1 + e)$$

$$= \sqrt{6g} \left(1 + \frac{1}{\sqrt{3}} \right) \text{ m/s}$$

PART-1_PHYSICS

SECTION-II

1. Ans (8.00)

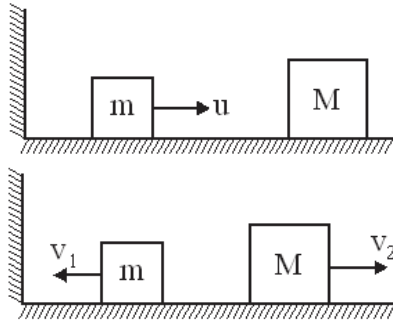
Liquid will be in pure translation

$$P_B - P_A = 2R\rho g,$$

$$K = \frac{1}{2} m V_0^2 + \frac{1}{2} \left(mR^2 + \frac{2}{3} mR^2 \right) \left(\frac{V_0}{R} \right)^2$$

$$\therefore \frac{K}{P_B - P_A} = \frac{8\pi R^2 V_0^2}{9g} \therefore n = 8$$

2. Ans (2.00)

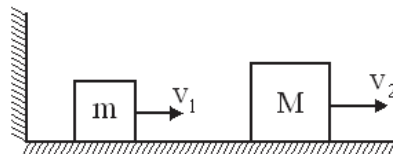


$$V_{cm} = \frac{mu}{m+M}$$

$$V_1 = -\frac{2mu}{m+M} + u$$

$$V_2 = \frac{2mu}{m+M}$$

For no collision, $V_2 \geq V_1$



$$\Rightarrow \frac{2mu}{m+M} \geq u - \frac{2mu}{m+M}$$

$$\Rightarrow \frac{4m}{m+M} \geq 1$$

$$\Rightarrow \boxed{3m \geq M}$$

3. Ans (5.00)

$$f = f_0 \frac{z^2}{n^3} \Rightarrow \frac{z^2}{n^3} = 2 \dots\dots\dots(1)$$

$$E = E_0 \frac{z^2}{n^2} \Rightarrow \frac{z^2}{n^2} = 4 \dots\dots\dots(2)$$

Solving we get $n = 2, z = 4$

$$L = mvr = \frac{nh}{2\pi}$$

$$\therefore \Delta L = \Delta n \frac{h}{2\pi}$$

$$\text{Also, } \Delta L = \tau \Delta t$$

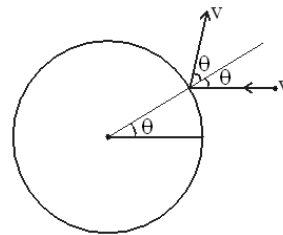
$$\therefore \tau \Delta t = \Delta n \cdot \frac{h}{2\pi} \Rightarrow \tau = \frac{\Delta n}{\Delta t} \frac{h}{2\pi}$$

$$\Rightarrow \tau = \frac{1}{7 \times 10^{-9}} \times \frac{2.1 \times 10^{-34}}{2} = 15 \times 10^{-27}$$

$$\therefore \frac{\tau}{3} \times 10^{27} = \frac{15 \times 10^{-27}}{3} \times 10^{27} = 5$$

4. Ans (8.00)

In the frame of the cylinder the particles come in with speed v and then bounce off with same speed as shown.



Force

$$F = \int_{-\pi/2}^{\pi/2} (2mv \cos \theta) \cos \theta \cdot (R d\theta \cos \theta) \ell v n$$

$$\Rightarrow \frac{F}{\ell} = \frac{8}{3} mv^2 n R \quad \therefore K = 8$$

5. Ans (2.00)

$$\vec{v} = \frac{v_0}{2} (\sqrt{3}\hat{i} + \hat{j})$$

For crossing origin,

$$K \frac{2\pi m}{qB} = \frac{2(v_0/2)}{(qE/m)}$$

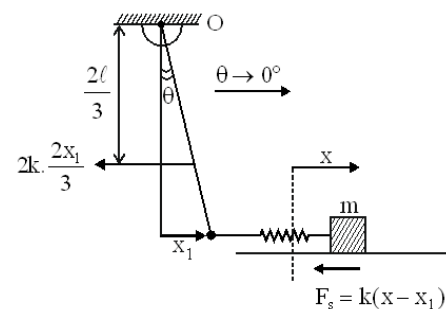
Where K has to be integer.

$$\Rightarrow B = \left(\frac{2\pi E}{v_0} \right) K$$

For minimum value $K = 1$

$$\Rightarrow B = \frac{2\pi E}{v_0}$$

6. Ans (17.00)



Light rod $\vec{\tau}_0 = 0$

$$2K \cdot \frac{2x_1}{3} \cdot \frac{2\ell}{3} = K(x - x_1) \cdot \ell$$

$$\Rightarrow \frac{17}{9} x_1 = x$$

Now for the block

$$K(x - x_1) = ma$$

$$\Rightarrow K \cdot \left(x - \frac{9x}{17} \right) = ma$$

$$\Rightarrow -\frac{8K}{17m} \vec{x} = \vec{a}$$

$$\text{Time period is } T = 2\pi \sqrt{\frac{17m}{8K}}$$

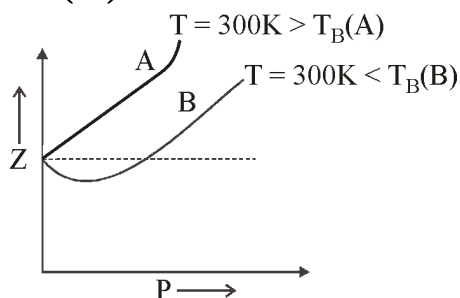
PART-2_CHEMISTRY

SECTION-I(i)

1. **Ans (C)**

C.N, density and distance between closest octahedral voids ($2r$) is same for both hcp and fcc unit cells.

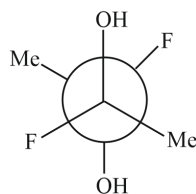
2. **Ans (C)**



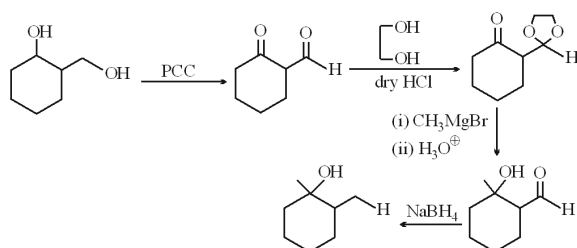
3. **Ans (A)**

I, II & III will have chiral gauche conformer as the most stable conformer.

IV- For the meso compound the most stable conformer is



4. **Ans (B)**



5. **Ans (C)**

Theory Based

6. **Ans (A)**

Theory Based.

PART-2_CHEMISTRY

SECTION-I(ii)

7. **Ans (A,B,C)**

$$(a) \Delta U = W$$

$$\Rightarrow \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} = -P_{\text{ext}}(V_2 - V_1)$$

$$\Rightarrow \frac{P_2 V_2 - 1 \times 20}{\frac{5}{3} - 1} = -P_2(V_2 - 20)$$

$$\Rightarrow 3(P_2 V_2 - 20) = 2(-P_2 V_2 + 20P_2)$$

$$\Rightarrow 5P_2 V_2 - 40P_2 = 60$$

$$\Rightarrow P_2 V_2 - 8P_2 = 12$$

$$\Rightarrow V_2 = \frac{12}{P_2} + 8$$

As $P_2 \rightarrow \infty$, $V_2 \rightarrow 8$ (lowest possible value)

$$(b) V_2 = \frac{12}{3} + 8 = 12L$$

$$\begin{aligned} \Delta S &= nC_V \ln \left(\frac{P_2}{P_1} \right) + nC_P \ln \left(\frac{V_2}{V_1} \right) \\ &= 1 \times \frac{3R}{2} \ln \left(\frac{3}{1} \right) + 1 \times \frac{5R}{2} \ln \left(\frac{12}{20} \right) \\ &= \left[3 \ln 3 + 5 \ln \left(\frac{3}{5} \right) \right] \text{ Cal K}^{-1} \\ &= 0.8 \text{ Cal K}^{-1} \end{aligned}$$

(c) For adiabatic process,

$$\Delta S_{\text{Surr}} = 0$$

$$\Delta S_{\text{sys}} = +ve$$

(d) $TV^{\gamma-1}$ or $TV^{(R/CV)}$ is only valid for reversible processes.

8. **Ans (B,D)**

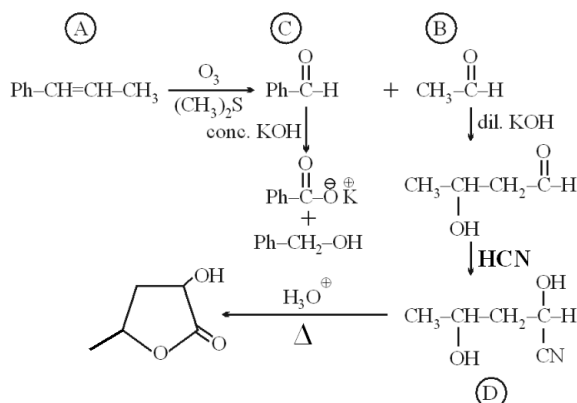
HNO_3 and H_2O form maximum boiling

azeotrope and upon distillation, the distillate

moves farther away from azeotrope composition,

i.e. $y < \chi$ for $\chi < \chi_{\text{azeotrope}}$

9. Ans (B,C,D)



11. Ans (A,D)

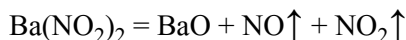
(A) Only Cl is ionisable.

(B) Fe^{+3} has odd number of electrons ($3d^5$), so paramagnetic.

(C) Only shows optical isomerism.

(D) As C_2H_4 bonds to Pt it's an organometallic compound.

12. Ans (A,B,C)

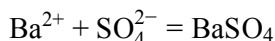


(A) (D)

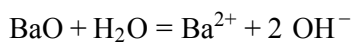
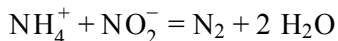
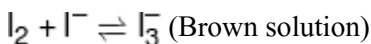
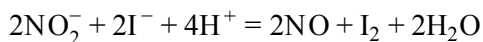


(B)

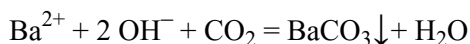
NH_4HSO_4 boils without further decomposition at 490°C .



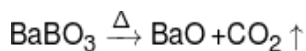
(C)



(D)



(E)



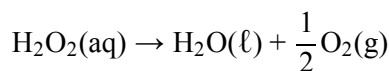
(E)

(D)

PART-2_CHEMISTRY

SECTION-II

1. Ans (3.00)



$\Delta H_r = \Delta H_{f\text{H}_2\text{O}}^\circ - \Delta H_{f\text{H}_2\text{O}_2}^\circ$

$= -287 - (-187) = -100 \text{ kJ/mol.}$

$\Rightarrow -\frac{d[\text{H}_2\text{O}_2]}{dt} = k[\text{H}_2\text{O}_2]$

$= 7.5 \times 10^{-4} \times 0.02 \text{ mol L}^{-1} \text{ s}^{-1}.$

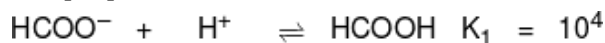
$\therefore$ Heat evolved per second

$= (2 \times 7.5 \times 10^{-4} \times 0.02 \times 100 \times 10^3) \text{ Js}^{-1} = 3 \text{ Js}^{-1}.$

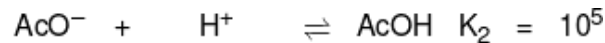
2. Ans (4.50)

Initial concentration of $[\text{HCOO}^-]$, $[\text{CH}_3\text{COO}^-]$

and $[\text{H}^+] = 1\text{M}$



$(1-x) \quad (1-x) \quad - \quad x$



$(1-y) \quad (1-x-y) \quad y$

$\therefore$ The equilibrium constants are very large, H^+ acts as the limiting reagent, and $1-x-y \approx 0$

$\Rightarrow x+y \approx 1 \dots(i)$

$K_1 = 10^4 = \frac{x}{(1-x)[\text{H}^+]}$

$\Rightarrow \frac{x}{1-x} = 10^4[\text{H}^+] \dots(ii)$

$K_2 = 10^5 = \frac{y}{(1-y)[\text{H}^+]}$

$\Rightarrow \frac{1-y}{y} = \frac{1}{10^5[\text{H}^+]}$

Also $y = 1-x$

$\Rightarrow \frac{x}{1-x} = \frac{1}{10^5[\text{H}^+]} \dots(iii)$

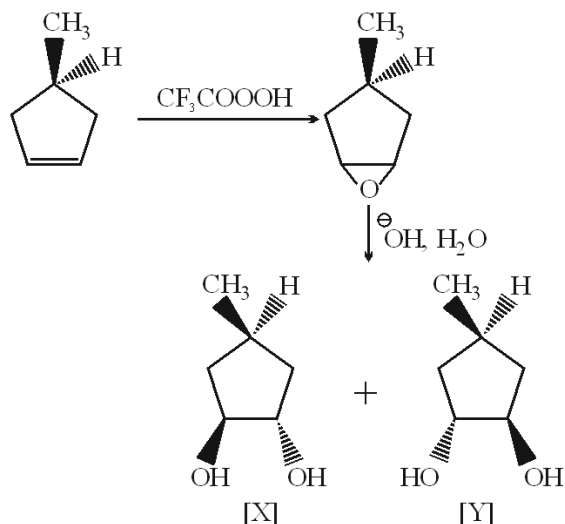
$(ii) = (iii) \Rightarrow 10^4[\text{H}^+] = \frac{1}{10^5[\text{H}^+]}$

$\Rightarrow [\text{H}^+]^2 = 10^{-9}$

$\Rightarrow \text{pH} = 4.5$

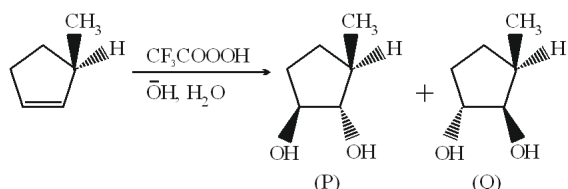
3. Ans (3.00)

(1)



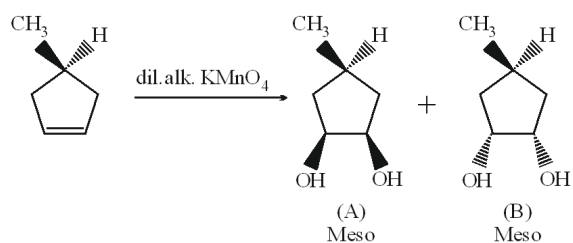
X and Y are enantiomer

(2)



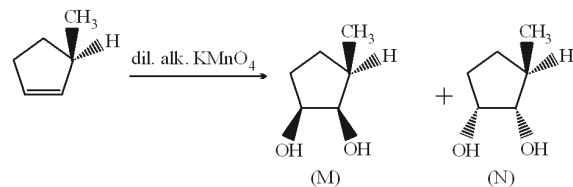
P and Q are diastereomer

(iii)



A and B are diastereomer

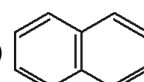
(iv)

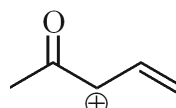


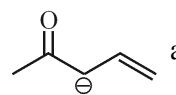
Correct statements are (1), (3) & (6)

4. Ans (6.00)

(1), (3) extended conjugation stabilizes more than cross conjugation

(2)  Both rings are aromatic

(4) In , cation is destabilized by $-\text{C}(=\text{O})-$ group.

(5) In , anion is stabilized by $-\text{C}(=\text{O})-$ group.

(6) Cation is not delocalised.

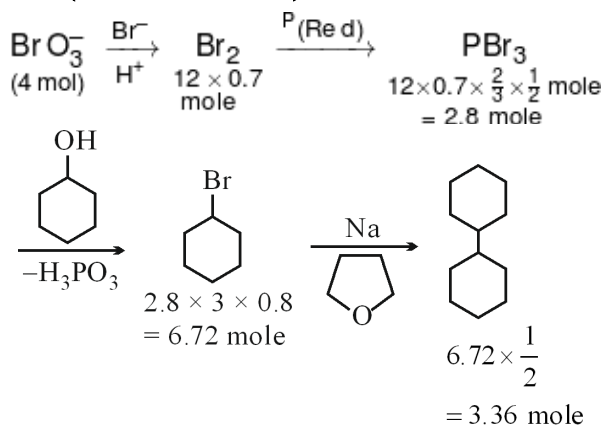
(7) 3° carbocation is more stable than 2° carbocation

(8) Uncharged resonating structure is more stable than charged.

(9) Cation is de-stabilised by -I effect.

Correct statements are 2,3,4,6,7, & 8.

5. Ans (556.10 to 557.80)



3.36 mole of D = $3.36 \times 166 = 557.76$

6. Ans (8.90 to 9.10)

dipole moment of HCHO = $0.4 + 2.3 = 2.7 \text{ D}$

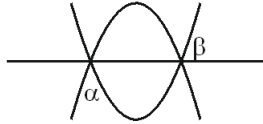
$2.7 \text{ D} = 2.7 \times 3.335 \times 10^{-30} \text{ C m} = 9.00 \times 10^{-30} \text{ C m}$

PART-3_MATHEMATICS

SECTION-I(i)

1. **Ans (D)**

$$(x^2 + 5x - b^2)(-2x^2 + ax + 6) = 0$$



$$\Rightarrow \text{both roots are in common } \frac{1}{-2} = \frac{5}{a} = \frac{-b^2}{6}$$

$$\Rightarrow a = -10, b^2 = 3$$

$$\text{Eqn. } x^2 + 5x - 3 = 0$$

$$\Rightarrow a + b = -5, \alpha\beta = -3$$

2. **Ans (C)**

Range of $f(x)$ contains $[0, \infty)$

$\therefore$ Range of $\text{gof}(x)$ is $(0, \pi]$

$\Rightarrow$ Number of integers are 3.

3. **Ans (D)**

$$\lim_{x \rightarrow 0^+} \frac{((1 + \tan x)^{\cot x} + e^{\cos x})((1 + \tan x)^{\cot x} - e^{\cos x})}{\frac{1 - \cos(\ln(1 + \sin \sqrt{x}))}{\ln^2(1 + \sin \sqrt{x})} \times \frac{\ln^2(1 + \sin \sqrt{x})}{(\sin \sqrt{x})^2} \times \frac{\sin^2 \sqrt{x}}{(\sqrt{x})^2} \times x}$$

$$4e \lim_{x \rightarrow 0^+} \frac{(1 + \tan x)^{\cot x} - e^{\cos x}}{x}$$

$$\Rightarrow 4e \lim_{x \rightarrow 0^+} \frac{e^{1 - \frac{1}{2}\tan x + \frac{1}{3}\tan^2 x + \dots} - e^{\cos x}}{x} =$$

$$\frac{e^{\cos x} \left[e^{\left(1 - \frac{1}{2}\tan x + \frac{1}{3}\tan^2 x + \dots\right) - \cos x} - 1 \right]}{x}$$

$$4e \lim_{x \rightarrow 0^+} \frac{x}{\left(1 - \frac{1}{2}\tan x + \frac{1}{3}\tan^2 x + \dots\right) - \cos x}$$

$$= 4e^2 \lim_{x \rightarrow 0^+} \frac{x}{\left(1 - \frac{1}{2}\tan x + \frac{1}{3}\tan^2 x + \dots\right) - \cos x}$$

$$= 4e^2 \left(-\frac{1}{2}\right) = -2e^2$$

4. **Ans (A)**

$$\int \left(e^x \left(\frac{\sin x + \cos x}{(\sin x + \cos x)^2} - \frac{\cos x - \sin x}{(\sin x + \cos x)^2} \right) + e^x \left(\frac{\sin x - \cos x}{(\sin x - \cos x)^2} - \frac{\cos x + \sin x}{(\sin x - \cos x)^2} \right) \right) dx$$

$$= e^x \left[\frac{\sin x + \cos x}{(\sin x + \cos x)^2} \right] + e^x \left[\frac{\sin x - \cos x}{(\sin x - \cos x)^2} \right] + C$$

$$= e^x \left[\frac{1}{\sin x + \cos x} + \frac{1}{\sin x - \cos x} \right]$$

$$= e^x \left[\frac{2 \sin x}{\sin^2 x - \cos^2 x} \right] + C$$

5. **Ans (B)**

$$S = (\sin^3 \theta + \sin^3 3\theta + \dots + \sin^3 19\theta) +$$

$$(\cos^3 \theta + \cos^3 3\theta + \dots + \cos^3 19\theta)$$

$$\text{Using, } \sin^3 \theta = \frac{3 \sin \theta - \sin 3\theta}{4} \quad \text{and}$$

$$\cos^3 \theta = \frac{\cos 3\theta + 3 \cos \theta}{4}$$

$$\text{Let } S_1 = (\sin^3 \theta + \sin^3 3\theta + \dots + \sin^3 19\theta)$$

$$= \frac{3}{4} (\sin \theta + \sin 3\theta + \dots + \sin 19\theta)$$

$$- \frac{1}{4} (\sin 3\theta + \sin 9\theta + \dots + \sin 57\theta)$$

$$= \frac{3}{4} \left(\frac{\sin 10\theta}{\sin \theta} \cdot \sin 10\theta \right)$$

$$- \frac{1}{4} \left(\frac{\sin 30\theta}{\sin 3\theta} \cdot \sin 30\theta \right)$$

$$\text{For } \theta = 15^\circ, \quad S_1 = \frac{3}{4} \left(\frac{1}{4} \cdot \frac{2\sqrt{2}}{\sqrt{3}-1} \right)$$

$$- \frac{1}{4} (\sqrt{2}) = \frac{3(\sqrt{3}+1)-4}{8\sqrt{2}}$$

$$\text{and let } S_2 = (\cos^3 \theta + \cos^3 3\theta + \dots + \cos^3 19\theta)$$

$$= \frac{3}{4} \left(\frac{\sin 10\theta}{\sin \theta} \cdot \cos 10\theta \right) + \frac{1}{4} \left(\frac{\sin 30\theta \cos 30\theta}{\sin 3\theta} \right)$$

For $\theta = 15^\circ$,

$$S_2 = \frac{3}{4} \left(\frac{-\sqrt{3}}{4} \cdot \frac{2\sqrt{2}}{\sqrt{3}-1} \right) + \frac{1}{4} (0)$$

$$= \frac{-3\sqrt{3}(\sqrt{3}+1)}{8\sqrt{2}}$$

$$S = S_1 + S_2 = \frac{3(\sqrt{3}+1)-4}{8\sqrt{2}} - \frac{3\sqrt{3}(\sqrt{3}+1)}{8\sqrt{2}}$$

$$= \frac{3\sqrt{3}-1-9-3\sqrt{3}}{8\sqrt{2}} = \frac{-10}{8\sqrt{2}} = \frac{-5}{\sqrt{32}} \equiv \frac{-a}{\sqrt{b}}$$

$$\therefore a = 5, b = 32$$

$$\Rightarrow |b - 6a| = 2.$$

6. Ans (B)

$\therefore f(x)$ is non-decreasing

$$\Rightarrow f'(x) \geq 0$$

$$\Rightarrow 3x^2 + 2ax + b \geq 0 \Rightarrow D \leq 0$$

$$\Rightarrow 4a^2 - 4.3.b \leq 0 \Rightarrow a^2 \leq 3b$$

$$a = 1, b = 1, 2, 3, 4, 5, 6$$

$$a = 2, b = 2, 3, 4, 5, 6$$

$$a = 3, b = 3, 4, 5, 6$$

$$a = 4, b = 6$$

$\Rightarrow a$ & b can vary in 16 ways

$\Rightarrow a, b, c$ can vary in $16 \times 6 = 96$ ways

$\therefore f(x)$ has all roots real, $f(x)$ must be of the form $(x + \alpha)^3$

only possible case $(x+1)^3 \Rightarrow a = 3, b = 3, c = 1$

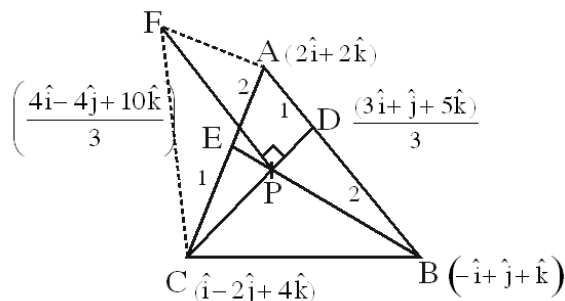
$$\Rightarrow \text{required probability} = \frac{1}{96}$$

PART-3_MATHEMATICS

SECTION-I(ii)

7. Ans (B,C,D)

$$CD; \vec{r} = (\hat{i} - 2\hat{j} + 4\hat{k}) + \lambda (7\hat{j} - 7\hat{k})$$



$$BE; \vec{r} = (-\hat{i} + \hat{j} + \hat{k}) + m(7\hat{i} - 7\hat{j} + 7\hat{k})$$

$\therefore$ Point of intersection of CD & BE is P

$$\therefore P = (\hat{i} - \hat{j} + 3\hat{k})$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 1 & -1 \\ -1 & -2 & 2 \end{vmatrix}$$

$$= \hat{i}(0) - \hat{j}(-7) + \hat{k}(7) = 7\hat{j} + 7\hat{k}$$

$$\vec{PB} \times \vec{PC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 2 & -2 \\ 0 & 1 & -1 \end{vmatrix}$$

$$= (-2)\hat{j} + (-2)\hat{k} = -2\hat{j} - 2\hat{k}$$

Area of tetrahedral ABCF =

$$\frac{1}{3}(\text{Base Area}) \times \text{Height} = \frac{1}{3} \left(\frac{7\sqrt{2}}{2} \right) \times \sqrt{2} = \frac{7}{3}$$

$$\vec{PF} = \pm \sqrt{2} \left(\frac{\hat{j} + \hat{k}}{\sqrt{2}} \right) = (\hat{j} + \hat{k}) = \vec{F} - \vec{P}$$

$$\therefore \vec{F} = (\hat{i} - \hat{j} + 3\hat{k}) \pm (\hat{j} + \hat{k}) = \hat{i} + 4\hat{k}; \hat{i} - 2\hat{j} + 2\hat{k}$$

$$AF; \vec{r} = 2(\hat{i} + \hat{k}) + \lambda(\hat{i} - 2\hat{k});$$

$$\vec{r} = 2(\hat{i} + \hat{k}) + \lambda(\hat{i} + 2\hat{j})$$

8. Ans (A,C,D)

Any point on the line $x + y = 1$ will be $(t, 1 - t)$.

Equation of chord of the ellipse with this point as a midpoint

$$\frac{tx}{2a^2} + \frac{y(1-t)}{2} = \frac{t^2}{2a^2} + \frac{(1-t)^2}{2}$$

it passes through $(a, -1)$ so that

$$\frac{at}{2a^2} + \frac{(t-1)}{2} = \frac{t^2}{2a^2} + \frac{(1-t)^2}{2}$$

$$at + a^2(t-1) = t^2 + a^2(1-t)^2$$

$$(a^2 + 1)t^2 - (3a^2 + a)t + 2a^2 = 0$$

Since we want two such chords

$$D > 0$$

$$(3a^2 + a)^2 - 4(a^2 + 1) \cdot 2a^2 > 0$$

$$(3a + 1)^2 - 8(a^2 + 1) > 0$$

$$a^2 + 6a - 7 > 0$$

$$(a + 7)(a - 1) > 0$$

$$a \in (-\infty, -7) \cup (1, \infty).$$

9. Ans (A,B,D)

$$f(x+y) = f(x) + f(y) + (x+y) \cdot e^{x+y} - x \cdot e^x - y \cdot e^y + 4xy$$

$$\text{Put } x = y = 0 \Rightarrow f(0) = 0$$

Differentiate partially w.r.t. x

$$f'(x+y) = f'(x) + (x+y) \cdot e^{x+y} + e^{x+y} - x \cdot e^x - e^x + 4y$$

$$\text{Put } y \rightarrow -x$$

$$f'(0) = f'(x) + 1 - x \cdot e^x - e^x - 4x$$

$$f'(x) = x \cdot e^x + e^x + 4x$$

$$f(x) = x \cdot e^x + 2x^2$$

10. Ans (C,D)

$$\int_0^{\pi} [\{x\} + [\cot x]] dx = \int_0^{\pi} ([\{x\}] + [\cot x]) dx$$

$$= \int_0^{\pi} [\cot x] dx$$

Now, $I = \int_0^{\pi} [\cot x] dx = \int_0^{\pi} [\cot (\pi - x)] dx$

$$2I = \int_0^{\pi} ([\cot x] + [-\cot x]) dx$$

$$2I = \int_0^{\pi} (-1) dx = -\pi \Rightarrow I = -\frac{\pi}{2}$$

11. Ans (A,B,C)

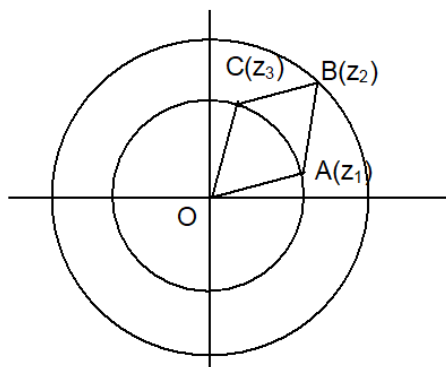
Given that $\text{Arg}(z_2 - z_1) = \frac{\pi}{3}$

since $\text{Arg } z_3 = \text{Arg}(z_2 - z_1) = \frac{\pi}{3}$

$$\Rightarrow z_3 = |z_3| e^{i\pi/3}$$

$$\Rightarrow z_3 = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$\Rightarrow z_3 = 2 \left(\frac{1}{2} + \frac{\sqrt{3}i}{2} \right) = (1 + \sqrt{3}i)$$



The mid point of diagonals $= \frac{z_1 + z_3}{2} = \frac{z_2 + 0}{2}$

$$\Rightarrow z_1 + z_3 = z_2 \quad \dots (1)$$

since $z_3 - z_1$ is perpendicular to z_2

$$\Rightarrow \frac{z_3 - z_1}{z_2} = ki \Rightarrow z_3 - z_1 = ki z_2 \quad (k \in \mathbb{R}) \quad \dots (2)$$

From (1) and (2) $z_1 = \left(\frac{1 - ki}{1 + ki} \right) z_3$ and $z_2 = \frac{2z_3}{1 + ki}$

$$\Rightarrow |z_2| = \frac{2|z_3|}{|1 + ki|} \Rightarrow 3 = \frac{2 \times 2}{\sqrt{1 + k^2}}$$

$$\Rightarrow 1 + k^2 = \frac{16}{9} \Rightarrow k = \pm \frac{\sqrt{7}}{3}$$

$$\Rightarrow z_2 = \left(\frac{2}{1 \pm \frac{\sqrt{7}}{3}i} \right) \cdot (1 + \sqrt{3}i)$$

$$\Rightarrow z_2 = \frac{3(2 + 2\sqrt{3}i)}{3 + \sqrt{7}i} \text{ or } \frac{3(2 + 2\sqrt{3}i)}{3 - \sqrt{7}i}$$

and $z_1 = \left(\frac{3 - \sqrt{7}i}{3 + \sqrt{7}i} \right) (1 + \sqrt{3}i)$ or $\left(\frac{3 + \sqrt{7}i}{3 - \sqrt{7}i} \right) (1 + \sqrt{3}i)$

As two such rhombuses are possible, hence one

set of values is

$$z_1 = \left(\frac{3 - \sqrt{7}i}{3 + \sqrt{7}i} \right) (1 + \sqrt{3}i)$$

$$z_2 = \frac{6 + 6\sqrt{3}i}{3 + \sqrt{7}i} \text{ and } z_3 = (1 + \sqrt{3}i) \text{ and another}$$

set of the values is

$$z_1 = \left(\frac{3 + \sqrt{7}i}{3 - \sqrt{7}i} \right) (1 + \sqrt{3}i),$$

$$z_2 = \left(\frac{6 + 6\sqrt{3}i}{3 - \sqrt{7}i} \right) \text{ and } z_3 = (1 + \sqrt{3}i).$$

12. Ans (B,C,D)

$$f'(x) = -3x^2 + 2px + q > 0 \quad \forall x \in (-2, 1)$$

Sum of roots $\Rightarrow \frac{2p}{3} = -1 \Rightarrow p = \frac{-3}{2}$

and product of roots $\Rightarrow \frac{-q}{3} = -2 \Rightarrow q = 6$

$$\therefore (p + q) = \frac{9}{2}.$$

PART-3_MATHEMATICS

SECTION-II

1. **Ans (9.50)**

$$\text{Normal vector of plane } \vec{n}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 3 & 2 \\ 2 & 3 & -1 \end{vmatrix} =$$

$$-9\hat{i} + 3\hat{j} - 9\hat{k}$$

$$\vec{a} \times \vec{b} = \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 6 & 3 \\ 2 & -6 & 5 \end{vmatrix} = 48\hat{i} - 4\hat{j} - 24\hat{k}$$

$\vec{a} = 2\hat{i} + 6\hat{j} + 3\hat{k}$
 $\vec{b} = 2\hat{i} - 6\hat{j} + 5\hat{k}$

$$\text{Required area} = \frac{1}{2} |\vec{a} \times \vec{b}| \hat{n}_2 \cdot \hat{n}_1$$

$$= \frac{1}{2} 4\sqrt{181} \cdot \frac{228}{4\sqrt{181}\sqrt{171}} = 2\sqrt{19} \equiv a\sqrt{b}$$

$$\therefore \left(\frac{b}{a}\right) = 9.50.$$

2. **Ans (119.00)**

Let page numbers $k, k+1, k+2, k+3$ are torn off.

$$\Rightarrow \frac{n(n+1)}{2} - (4k+6) = 1596$$

$$\Rightarrow n^2 + n - 4(2k+3) = 3192$$

$$\Rightarrow n^2 + n - 3192 = 4(2k+3)$$

$$\Rightarrow (n+57)(n-56) = 4(2k+3)$$

$$\text{if } n = 59, k = 42$$

$$\text{if } n = 60, k = 57$$

$$\text{Sum} = 59 + 60 = 119$$

3. **Ans (0.50)**

Let $A \equiv (\alpha, \beta)$ the equation of the normal to the parabola at $(at^2, 2at)$ is

$$y + tx = 2at + at^3$$

which passes through (α, β)

$$\text{So, } at^3 + (2a - \alpha)t - \beta = 0$$

$$\Rightarrow t_1 + t_2 + t_3 = 0 \quad \dots(1)$$

So, we have

$$P(at_1^2, 2at_1), Q \equiv (at_2^2, 2at_2)$$

$$\text{and } R \equiv (at_3^2, 2at_3)$$

$C_1 \equiv$ Circle drawn on AP as diameter

$$= (x - \alpha)(x - at_1^2) + (y - \beta)(y - 2at_1) = 0.$$

$C_2 \equiv$ Circle drawn on AQ as diameter

$$= (x - \alpha)(x - at_2^2) + (y - \beta)(y - 2at_2) = 0.$$

$$\text{Common chord} \equiv (t_1 + t_2)x + 2y - \alpha(t_1 + t_2) - 2\beta = 0$$

(as $t_1 \neq t_2$)

$$\therefore m_1 = -\left(\frac{t_1 + t_2}{2}\right) = \frac{t_3}{2} \quad (\text{from (1)})$$

Tangent to the parabola at R

$$t_3y = x + at_3^2 \Rightarrow m_2 = \frac{1}{t_3}$$

$$\text{so, } m_1 \times m_2 = \frac{1}{2}.$$

4. **Ans (1206.00)**

$$\text{Case-1: } 4, 1, 0, 0, 0, 0, 0, 0 \rightarrow \frac{9!}{7!} = 72$$

$$\text{Case-2: } 2, 2, 2, 2, 1, 0, 0, 0 \rightarrow \frac{9!}{4! \cdot 4!} = 630$$

$$\text{Case-3: } 2, 2, 2, 1, 1, 1, 1, 0 \rightarrow \frac{9!}{3! \cdot 5!} = 504$$

$$\therefore \text{Number of matrices in } S = 72 + 630 + 504 = 1206.$$

5. **Ans (19.20)**

$$\int_0^1 (6x^4 \cdot f(x) - f^2(x) - 9x^8 + 9x^8) dx = 1$$

$$\int_0^1 (9x^8 - (f(x) - 3x^4)^2) dx = 1$$

$$1 - \int_0^1 (f(x) - 3x^4)^2 dx = 1$$

$$\int_0^1 (f(x) - 3x^4)^2 dx = 0$$

$$\therefore f(x) = 3x^4$$

$$\therefore \text{Required Area} = \int_0^2 (3x^4) dx = \frac{3}{5} x^5 \Big|_0^2$$

$$= \frac{3}{5} \times (32 - 0) = 19.20.$$

 6. **Ans (5.00)**

$$u = \sin^7 x, v = \cos x$$

$$\frac{du}{dv} = \frac{7\sin^6 x \cos x}{-\sin x} = -7\sin^5 x \cos x$$

$$\frac{d^2u}{dv^2} = 7[5\sin^3 x \cos^2 x - \sin^5 x]$$

$$\frac{d^3u}{dv^3} = -\frac{105}{4} \sin 4x$$

$$\text{Let } \frac{d^3u}{dv^3} = z \Rightarrow z = \frac{-105}{4} \sin 4x$$

$$\Rightarrow \frac{dz}{dx} = \frac{-105}{4} (4 \cos 4x)$$

$$\Rightarrow \frac{d^2z}{dx^2} = -105 (-4 \sin 4x)$$

$$\frac{d^2z}{dx^2} \Big|_{x=\frac{\pi}{8}} = 105 \times 4 = 2^2 \cdot 3^1 \cdot 5^1 \cdot 7^1 \equiv$$

$$2^{\lambda_1} \cdot 3^{\lambda_2} \cdot 5^{\lambda_3} \cdot 7^{\lambda_4}$$

$$\Rightarrow \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 5.$$